

Product Vision Document: Scalable Event Ticketing Platform

Prepared for: Software Engineering II Project

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1. Executive Summary

This document outlines the product vision for a highly available, distributed, and scalable online event ticketing platform. The core objective is to deliver a fair, transparent, and resilient solution that thrives under high-concurrency traffic spikes during popular event releases, entirely eliminating double-booking and system crashes. The platform aims to bridge the gap between high-demand event launches and technical stability, ensuring that every user has an equal and fair opportunity to secure tickets without experiencing technical friction.

2. Long-Term Platform Goals & Strategic Objectives

- **Seamless User Experience for Discovery and Booking:** Ensure friction-free discovery, dynamic pricing updates, and instant seat availability visualization. The UI must reflect live venue layouts accurately.
- **Uncompromising Reliability and Concurrency Mitigation:** Guarantee fair distribution through robust concurrency control. We must implement advanced synchronization strategies to prevent double-booking and gracefully manage flash traffic spikes.
- **Reliable Transaction Management:** Deploy a bullet-proof checkout workflow using distributed transaction patterns (like Saga or compensation patterns) to securely process payments and issue tickets immediately, with automated rollbacks for failed transactions.
- **De-coupled Architecture for Independent Scaling:** Maintain a decoupled, microservices-based architecture (Identity, Catalog, Reservation, Billing, Notifications) capable of scaling independent domains based on real-time demand.

3. Targeted User Personas & Pain Points

Persona	Needs & Pain Points	Platform Solutions
The Enthusiastic Buyer (End-User)	Frustrated by website crashes during ticket drops, losing seats while paying, and clunky interfaces. Needs fairness and speed.	Virtual waiting rooms, temporary seat locks (TTL) using in-memory stores, real-time map updates via WebSockets, and immediate ticket issuance.
The Event Organizer	Lacks visibility into live sales, struggles with complex venue setups, fears technical failures ruining their brand reputation.	Real-time analytics dashboards, intuitive venue/sector management (rows, sections), automated capacity mapping, and dynamic pricing models.
Platform Administrator / DevOps	Needs to monitor platform health, mitigate malicious bot traffic, manage overarching access controls, and ensure 99.99% uptime.	Centralized Identity & Access Management (IAM), traffic throttling, comprehensive system telemetry (Prometheus/Grafana), and Kubernetes auto-scaling.
Financial & Data Analyst	Needs exact tracking of all successful, failed, and rolled-back transactions to ensure market trust and correct accounting.	Immutable transactional logs, clear saga state management, and strict relational database ACID compliance for the billing domain.

4. Core Business Metrics (KPIs)

To ensure alignment with market needs and operational excellence, the following metrics will guide development priorities:

- **Peak Concurrency Limit:** Measuring the number of simultaneous active users (bots and real users) the system can handle without response degradation. Target: Zero downtime during flash sales.
- **Transaction Success Rate:** Tracking the percentage of initiated checkouts that result in successful, verified ticket issuance versus timeouts or failures.

- **Seat Lock Contention Rate:** Measuring how many users are rejected smoothly when attempting to lock an already-locked seat, validating the concurrency engine.
- **Booking Abandonment Rate:** Analyzing drop-offs during the seat-lock phase to optimize the payment gateway integration.
- **System Uptime & Latency:** Aiming for 99.99% availability and tracking message broker dispatch latencies for instant notifications.

5. Technical Vision & Strategic Alignment

The engineering strategy is deeply tied to the business value, ensuring the architecture directly supports the product goals:

1. **High-Performance Concurrency:** By leveraging high-performance programming languages for concurrency management and in-memory data storage, the platform will handle intensive seat-locking operations effortlessly, preventing race conditions.
2. **Containerized Auto-Scaling:** A containerized infrastructure allows for dynamic scaling. When the virtual waiting room detects a traffic surge, the orchestrator will automatically spin up new instances of the reservation engine to absorb the load.
3. **Domain-Driven Design (DDD):** The system is strictly divided into isolated contexts (Identity, Catalog, Reservation, Billing, Notifications). This ensures that as the business expands, a failure in the Notification service does not crash the core Reservation engine.
4. **Asynchronous Event-Driven Architecture:** Utilizing message brokers buffers unexpected traffic bursts and ensures reliable delivery of order states from billing workers to notification triggers.